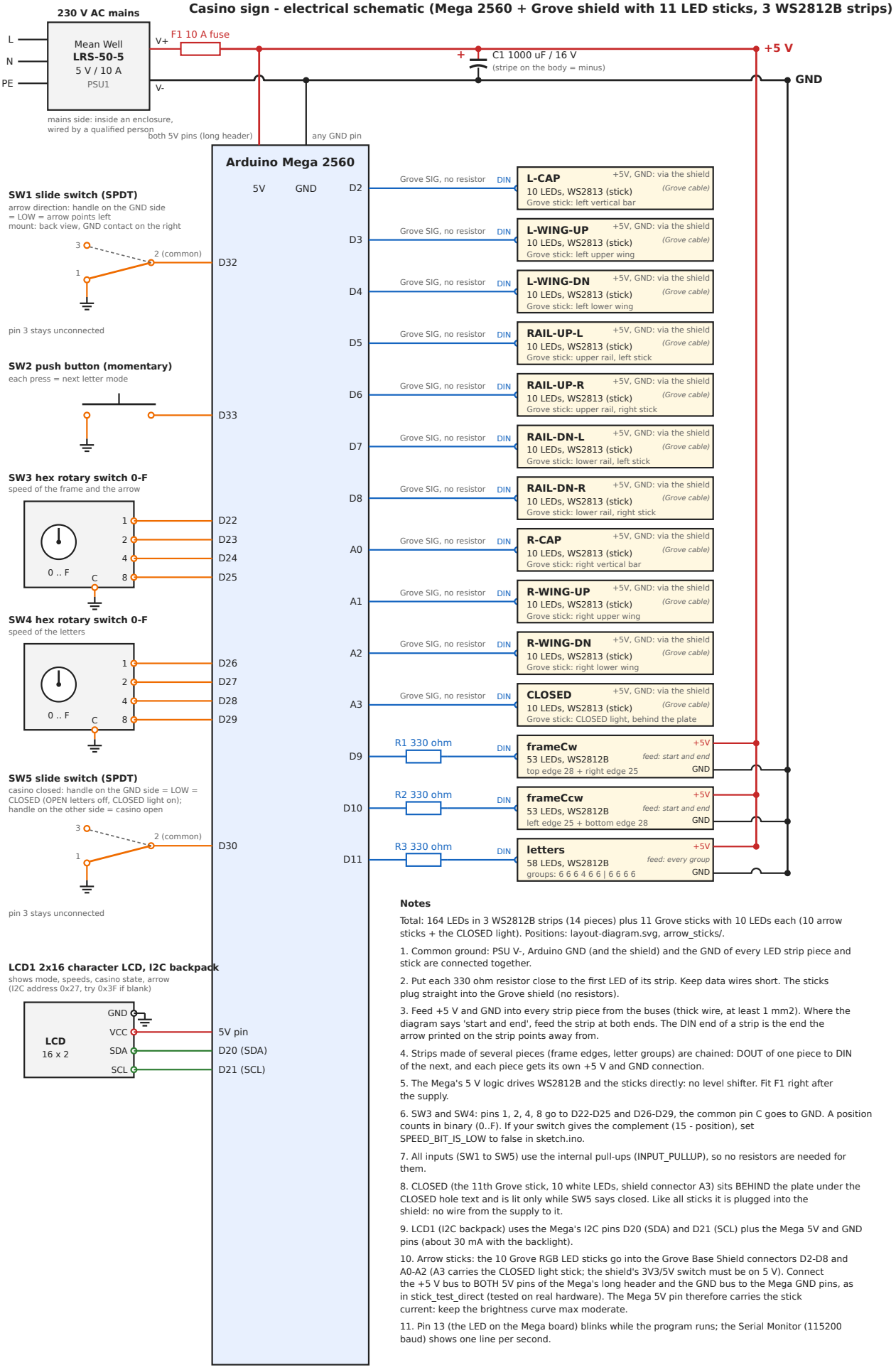
Casino sign - electrical schematic (Mega 2560 + Grove shield with 11 LED sticks, 3 WS2812B strips)



Notes

Total: 164 LEDs in 3 WS2812B strips (14 pieces) plus 11 Grove sticks with 10 LEDs each (10 arrow sticks + the CLOSED light). Positions: layout-diagram.svg, arrow_sticks/.

1. Common ground: PSU V-, Arduino GND (and the shield) and the GND of every LED strip piece and stick are connected together.

2. Put each 330 ohm resistor close to the first LED of its strip. Keep data wires short. The sticks plug straight into the Grove shield (no resistors).

3. Feed +5 V and GND into every strip piece from the buses (thick wire, at least 1 mm2). Where the diagram says 'start and end', feed the strip at both ends. The DIN end of a strip is the end the arrow printed on the strip points away from.

4. Strips made of several pieces (frame edges, letter groups) are chained: DOUT of one piece to DIN of the next, and each piece gets its own +5 V and GND connection.

5. The Mega's 5 V logic drives WS2812B and the sticks directly: no level shifter. Fit F1 right after the supply.

6. SW3 and SW4: pins 1, 2, 4, 8 go to D22-D25 and D26-D29, the common pin C goes to GND. A position counts in binary (0..F). If your switch gives the complement (15 - position), set SPEED_BIT_IS_LOW to false in sketch.ino.

7. All inputs (SW1 to SW5) use the internal pull-ups (INPUT_PULLUP), so no resistors are needed for them.

8. CLOSED (the 11th Grove stick, 10 white LEDs, shield connector A3) sits BEHIND the plate under the CLOSED hole text and is lit only while SW5 says closed. Like all sticks it is plugged into the shield: no wire from the supply to it.

9. LCD1 (I2C backpack) uses the Mega's I2C pins D20 (SDA) and D21 (SCL) plus the Mega 5V and GND pins (about 30 mA with the backlight).

10. Arrow sticks: the 10 Grove RGB LED sticks go into the Grove Base Shield connectors D2-D8 and A0-A2 (A3 carries the CLOSED light stick; the shield's 3V3/5V switch must be on 5 V). Connect the +5 V bus to BOTH 5V pins of the Mega's long header and the GND bus to the Mega GND pins, as in stick_test_direct (tested on real hardware). The Mega 5V pin therefore carries the stick current: keep the brightness curve max moderate.

11. Pin 13 (the LED on the Mega board) blinks while the program runs; the Serial Monitor (115200 baud) shows one line per second.

USB-B cable (upload, Serial Monitor): ONLY while the LED supply is switched OFF. The supply feeds the Mega 5V pins (the sticks get their power through the Grove shield), so with USB plugged in as well two 5 V sources would be connected together.

+5 V power — GND — data (via 330 ohm) — inputs — I2C — hop = lines cross without a connection; dot = connection